

Glucose Biosynthesis: Gluconeogenesis

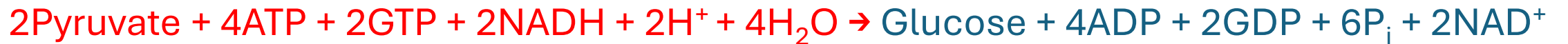
Gluconeogenesis

- Metabolic pathway that synthesizes glucose from non-carbohydrate precursors.
- **Purpose:** To maintain normal blood glucose levels, particularly during prolonged fasting, starvation, intense exercise, or when dietary carbohydrate intake is insufficient.

This is vital for tissues like the brain and red blood cells that heavily rely on glucose.
- **Location:** Primarily occurs in the **liver** and to a lesser extent in the **kidneys**.

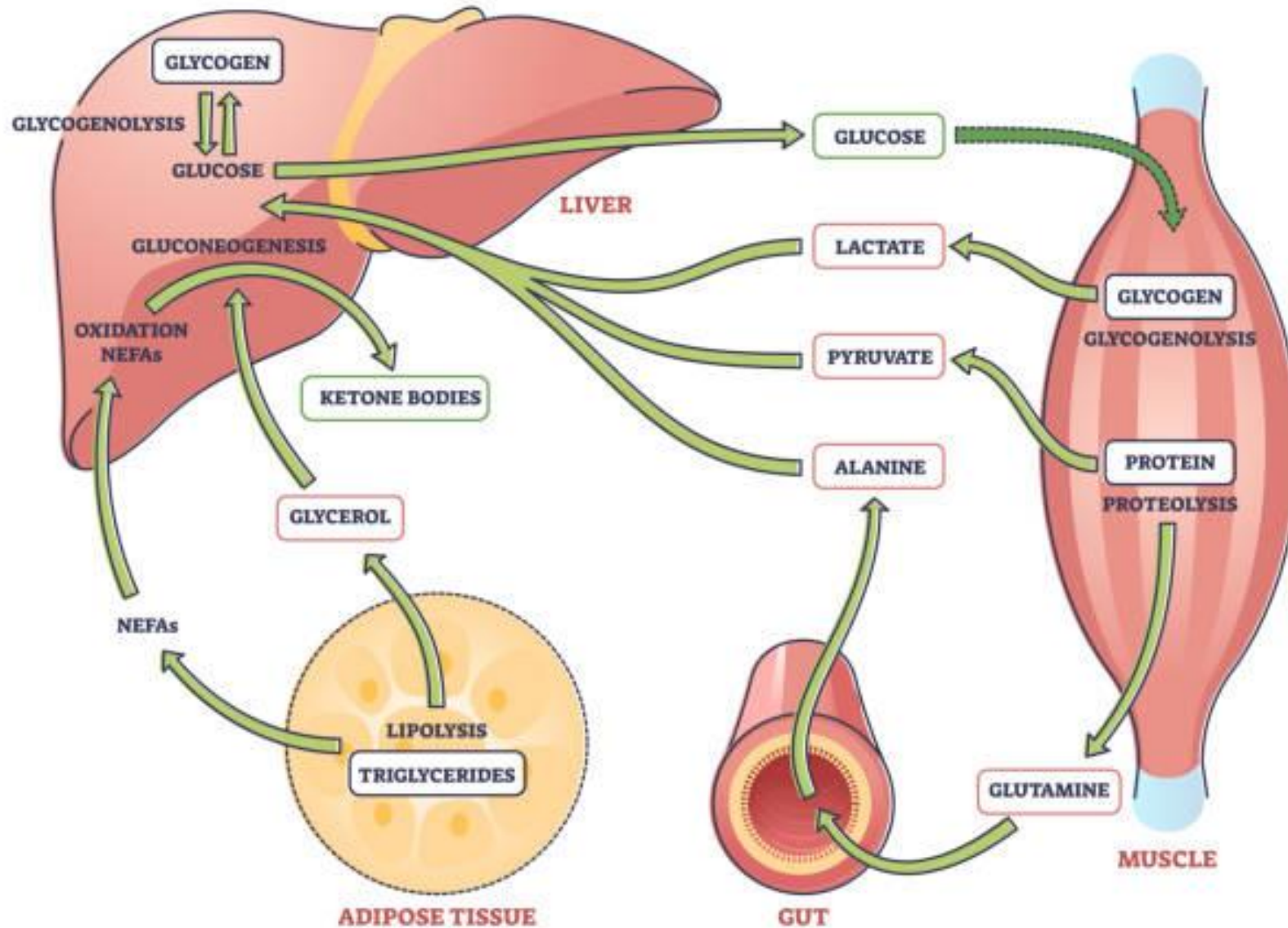
Gluconeogenesis

Overall Reaction:



- **Note the energy cost:** Gluconeogenesis is an energy-expensive process, consuming 6 ATP equivalents per glucose molecule synthesized. This ensures it doesn't run uncontrolled.

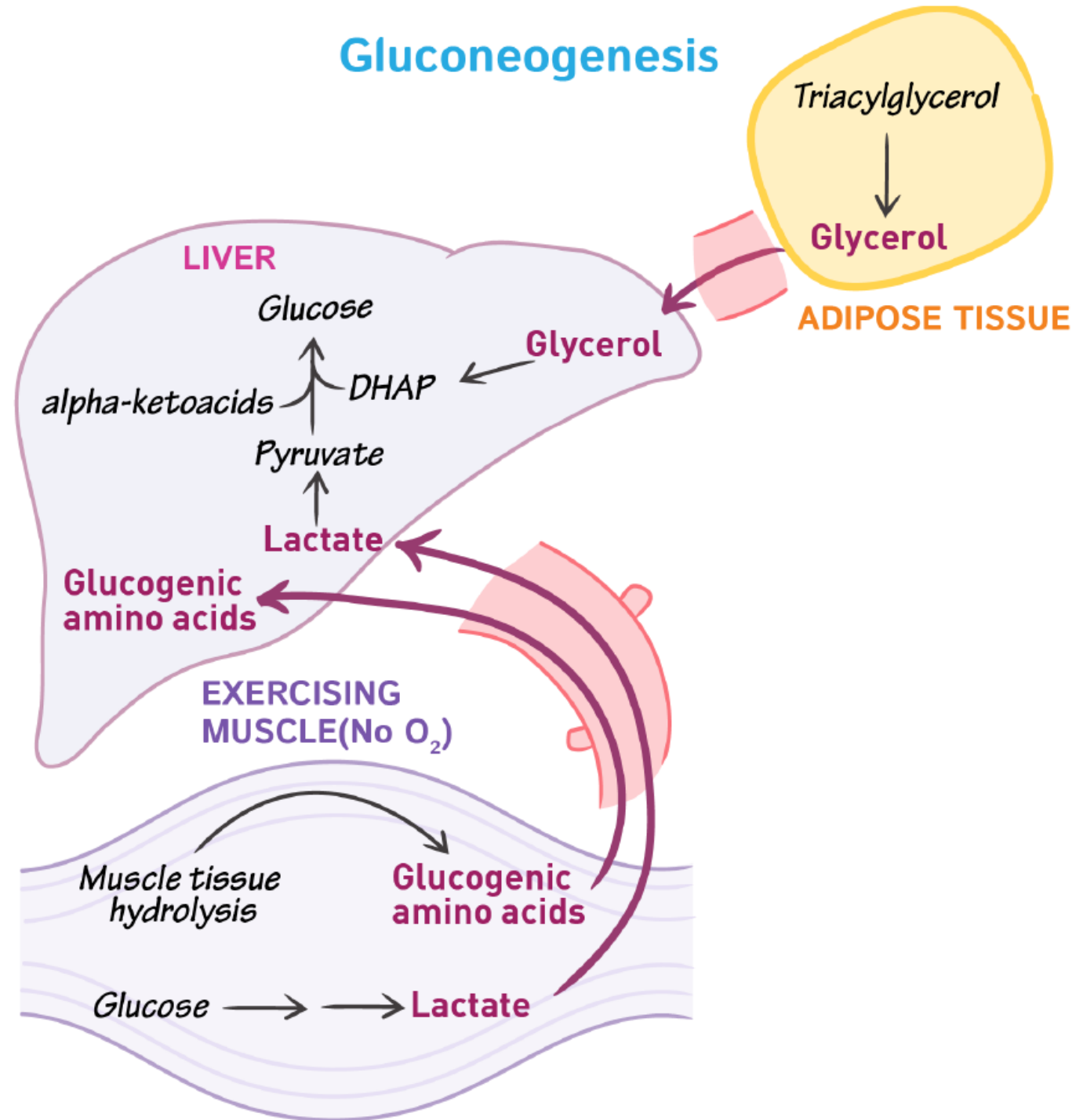
GLUCONEOGENESIS



Key Precursors

- **Lactate:** Produced by anaerobic glycolysis in exercising muscle and red blood cells. It's transported to the liver and converted back to pyruvate (Cori Cycle).
- **Amino Acids:** Most amino acids (except leucine and lysine) are "glucogenic," meaning their carbon skeletons can be converted to pyruvate or citric acid cycle intermediates, which can then be used for gluconeogenesis.
- **Glycerol:** Released from the breakdown of triglycerides (fats) in adipose tissue. It can be converted to DHAP, an intermediate of gluconeogenesis.
- **Important Note:** In animals, **fatty acids generally cannot be converted to glucose**. This is because the acetyl-CoA derived from fatty acid breakdown cannot be net converted to oxaloacetate (a key gluconeogenic intermediate) in the citric acid cycle.

Gluconeogenesis



Bypass Reactions (Overcoming Irreversible Glycolysis Steps)

- Glycolysis has three highly exergonic (irreversible) steps that must be bypassed in gluconeogenesis. These bypasses use different enzymes and often have different cellular locations.
 - **Bypass of Pyruvate Kinase (Glycolysis Step 10): Pyruvate → PEP**
 - **Bypass of PFK-1 (Glycolysis Step 3): Fructose-1,6-bisphosphate → Fructose-6-phosphate**
 - **Bypass of Hexokinase/Glucokinase (Glycolysis Step 1): Glucose-6-phosphate → Glucose**

Reactions of Gluconeogenesis

- 3 irreversible reactions of glycolysis bypassed by unique gluconeogenic enzymes
- 7 reversible reactions of glycolysis shared

BYPASS 1:

*Pyruvate
Kinase*

Phosphoenolpyruvate $\longrightarrow$ Pyruvate

BYPASS 2:

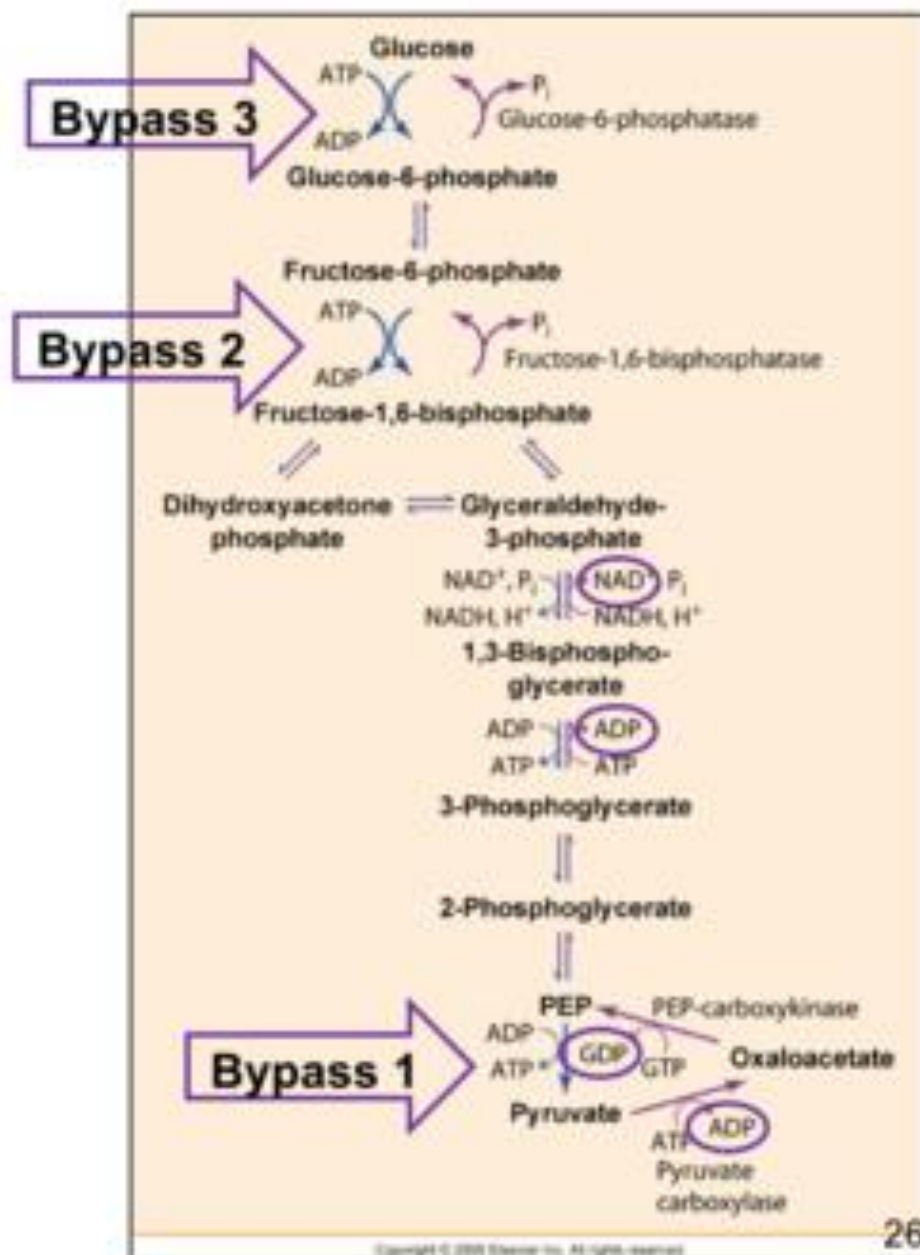
PFK-1

Fructose-6-P $\longrightarrow$ Fructose-1,6-BP

BYPASS 3:

Glucokinase

Glucose $\longrightarrow$ Glucose-6P



Regulation of Gluconeogenesis

Reciprocal Regulation with Glycolysis

- To prevent a futile cycle, the key regulatory enzymes of gluconeogenesis are regulated reciprocally to those of glycolysis.
- When glycolysis is active, gluconeogenesis is inhibited, and vice versa.

Regulation of Gluconeogenesis

Hormonal Control

- **Glucagon:** Released when blood glucose is low. It *stimulates* gluconeogenesis (and glycogenolysis) in the liver by promoting gene expression of gluconeogenic enzymes (e.g., PEPCK) and by phosphorylating (activating) FBPase-2, which lowers Fructose-2,6-bisphosphate levels, thus removing the inhibition on FBPase-1.
- **Cortisol:** A steroid hormone, also stimulates gluconeogenesis during prolonged stress.
- **Insulin:** Released when blood glucose is high. It *inhibits* gluconeogenesis by promoting dephosphorylation of enzymes (like FBPase-2, which increases Fructose-2,6-bisphosphate) and by reducing the expression of gluconeogenic enzymes.

Glycogen Metabolism: Biosynthesis and Degradation

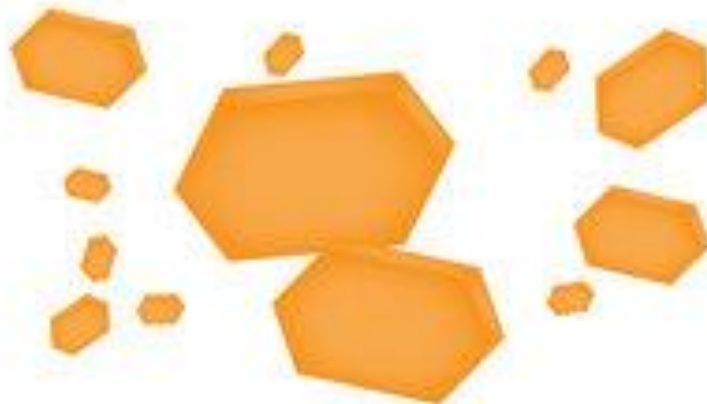
The liver stores
glucose as glycogen



Glycogenesis



Glycogenolysis



Glucose



Glycogenesis

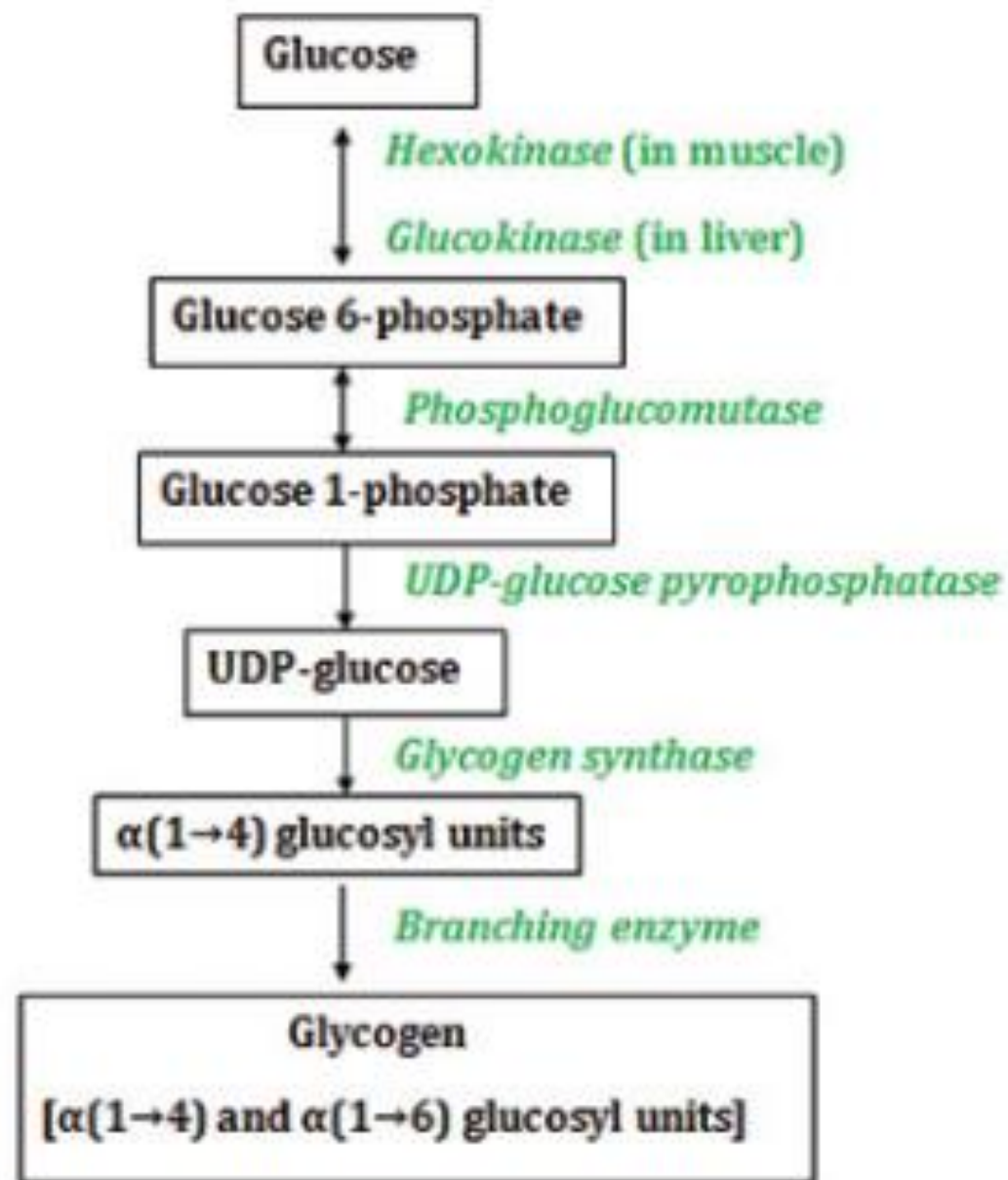
- Glycogen is the primary storage form of glucose in animals, providing a readily mobilizable energy reserve.

Glycogen Biosynthesis:

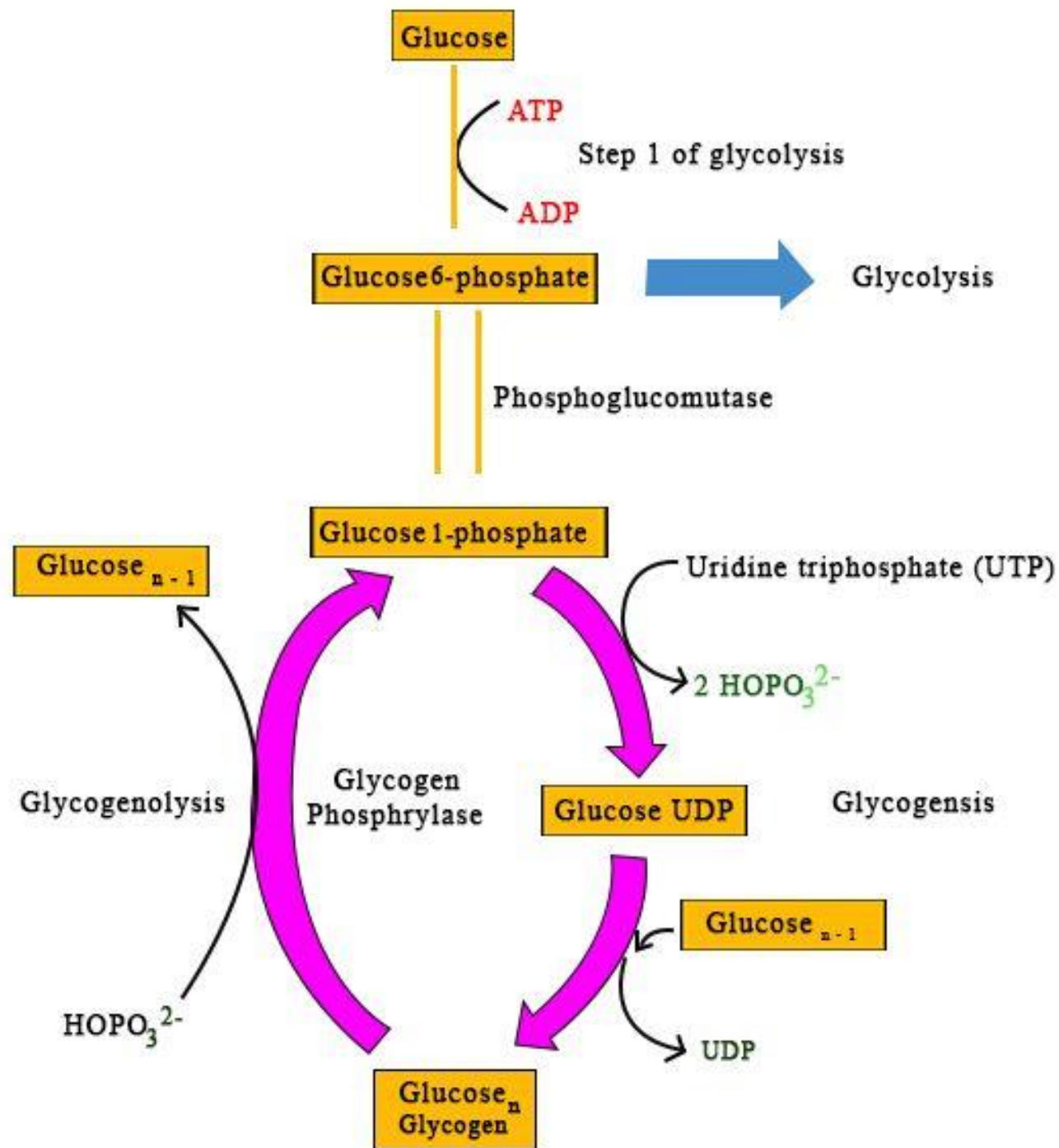
- **Purpose:** To synthesize glycogen from glucose, primarily when blood glucose levels are high (e.g., after a carbohydrate-rich meal). This stores excess glucose for later use.
- **Location:** Occurs in the cytosol of most cells, but is most prominent in the **liver** (stores glycogen for maintaining blood glucose) and **skeletal muscle** (stores glycogen for its own energy needs during activity).

Key Steps

- **Glucose Phosphorylation**
- **Isomerization**
- **Activation of Glucose**
- **Initiation (Glycogen Primer)**
- **Chain Elongation**
- **Branch Formation**



Glycogenesis



Regulation of Glycogen Synthase

Hormonal Control (Phosphorylation/Dephosphorylation)

- **Insulin:** (High blood glucose) Activates protein phosphatases, which dephosphorylate glycogen synthase, leading to its **activation**. This promotes glucose storage.
- **Glucagon (liver) / Epinephrine (muscle & liver):** (Low blood glucose/stress) Activate protein kinases (e.g., PKA), which phosphorylate glycogen synthase, leading to its **inactivation**. This prevents glucose storage when glucose is needed.

Allosteric Regulation

- **Activated by:** Glucose-6-phosphate (an allosteric activator, signaling high cellular glucose levels).

Glycogen Degradation: Glycogenolysis

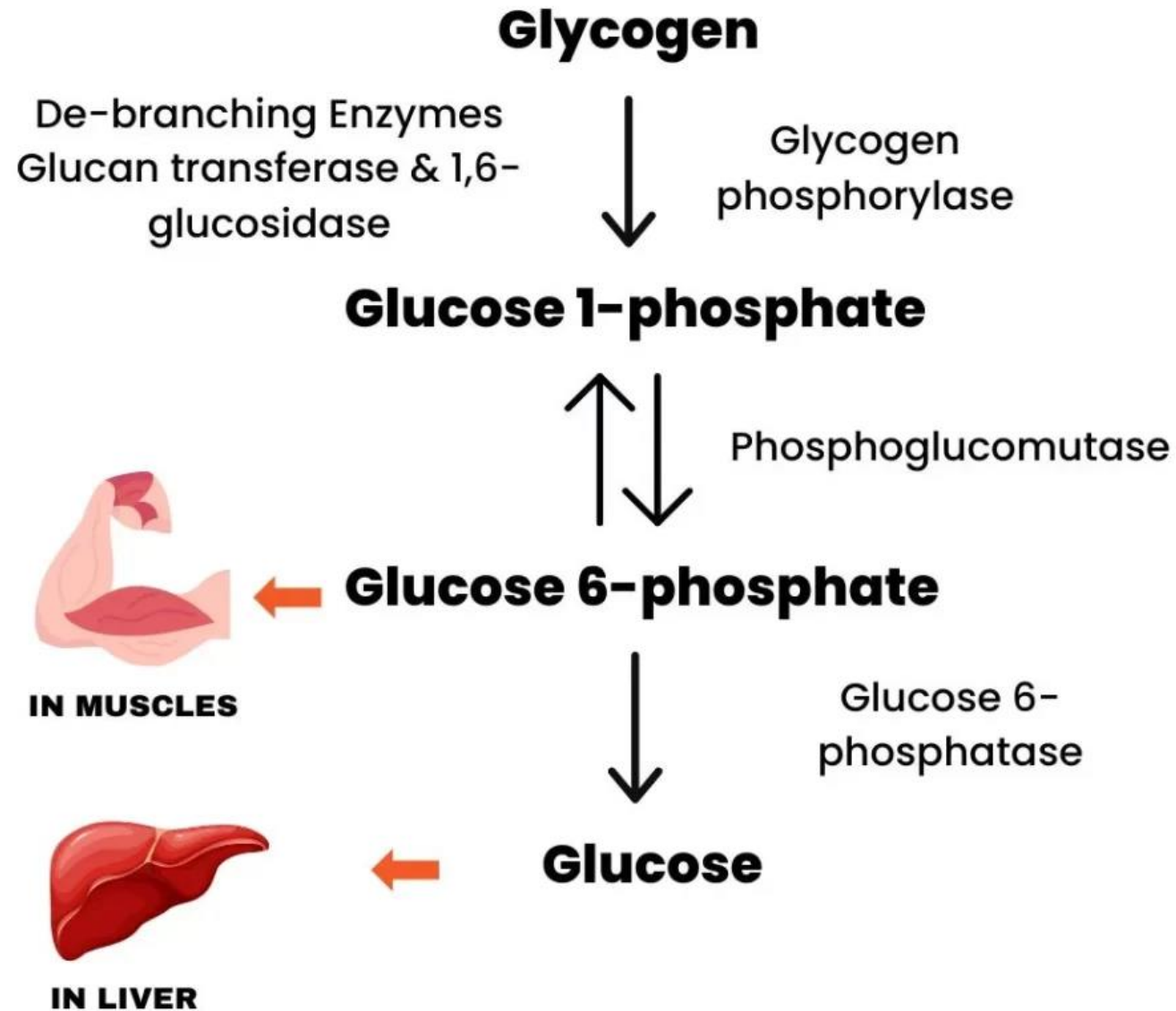
Glycogen Degradation: Glycogenolysis

- **Purpose:** To break down glycogen into glucose (or glucose-6-phosphate) to provide energy when glucose is scarce (e.g., during fasting) or when energy demands are high (e.g., during intense muscle activity).
- **Location:** Occurs in the cytosol of liver and muscle cells.

Key Steps

- **Phosphorolysis of $\alpha(1\rightarrow4)$ bonds**
- **Debranching**
- **Conversion of G1P to G6P**
- **Fate of Glucose-6-phosphate (G6P)**
 - **In Muscle:** Muscle cells lack Glucose-6-phosphatase. Therefore, G6P enters glycolysis directly to provide ATP for muscle contraction. Muscle glycogen only serves the muscle itself.
 - **In Liver:** The liver possesses **Glucose-6-phosphatase** (located in the endoplasmic reticulum). This enzyme dephosphorylates G6P, releasing **free glucose** into the bloodstream to maintain blood glucose homeostasis for the entire body.

STEPS OF GLYCOGENOLYSIS



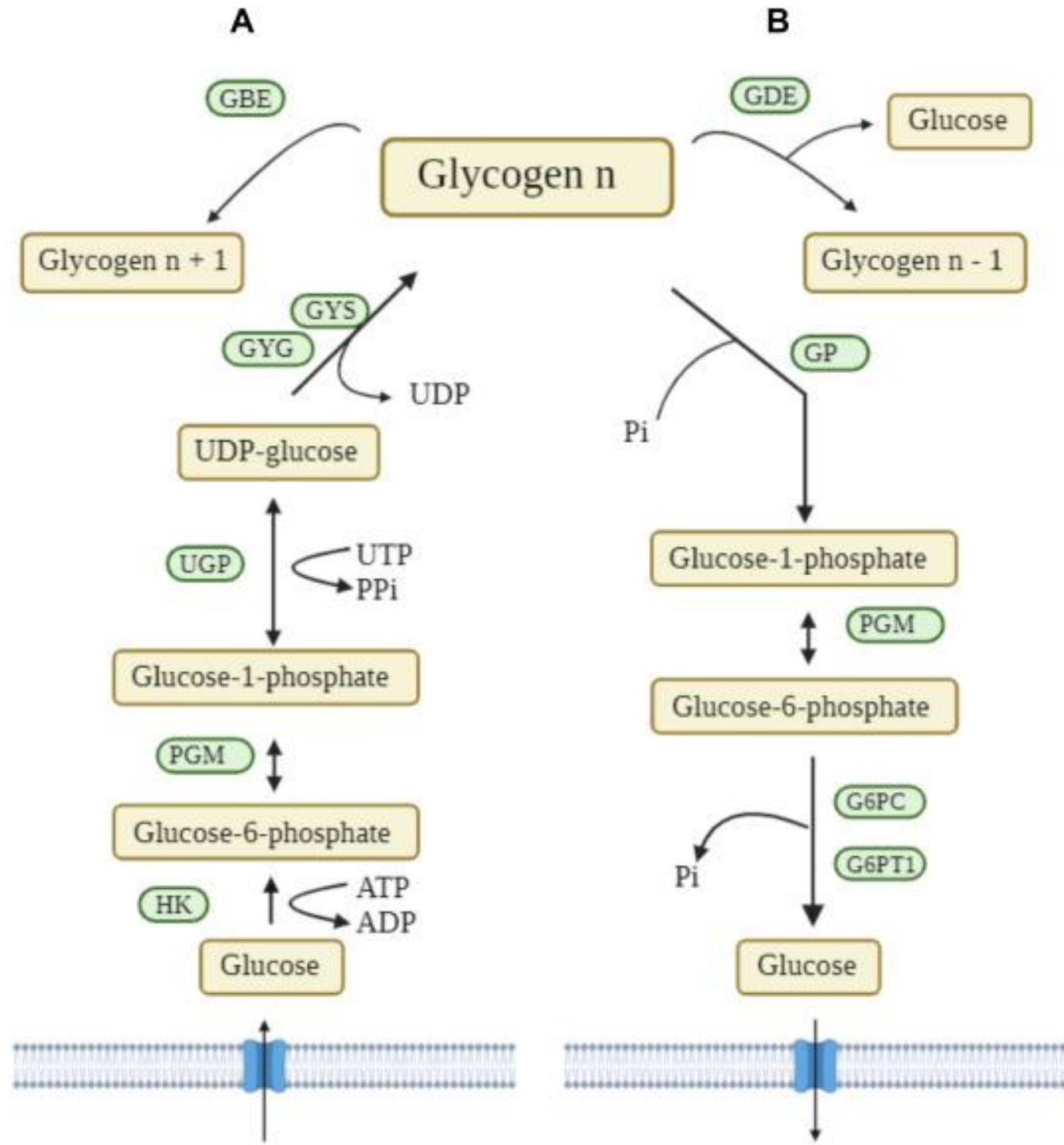
Regulation of Glycogen Phosphorylase

Hormonal Control (Phosphorylation/Dephosphorylation):

- **Glucagon (liver) / Epinephrine (muscle & liver):** (Low blood glucose/stress) Activate protein kinases (e.g., PKA), which phosphorylate glycogen phosphorylase, leading to its **activation**. This promotes glucose release.
- **Insulin:** (High blood glucose) Activates protein phosphatases, which dephosphorylate glycogen phosphorylase, leading to its **inactivation**.

Allosteric Regulation:

- **Activated by:** AMP (in muscle, signaling low energy charge, need for glucose). Calcium ions (Ca^{2+}) (in muscle, signaling muscle contraction and need for energy).
- **Inhibited by:** ATP, Glucose-6-phosphate (in muscle), and free glucose (in liver, signals high blood glucose).



THANK YOU